SE 216 – SOFTWARE PROJECT MANAGEMENT
REQUIREMENTS DOCUMENT

PROJECT NAME: Vaccine Integrity Monitoring System (VIMS)
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REQ. #	FUNCTIONAL REQUIREMENTS
1	The system must allow users to scan QR codes at every handoff point.
2	The system must collect temperature, humidity, and movement data from sensors at fixed intervals.
3	The system must transmit the collected sensor data to the central server in real time.
4	The system must generate an alert when temperature values fall outside the safe range.
5	The system shall retrieve environmental sensor data from the smart vaccine storage case.
6	The system shall predict potential cold chain failures by analyzing environmental trends and anomalies.
7	The system shall notify supervisors via mobile application and email when temperature deviations or connection losses are detected.
8	The system must allow authorized users to track the current location and status of the vaccine shipment.
9	The system shall generate a digital Arrival Report including temperature logs, humidity levels, timestamped location data, and any alert triggers occurred during the shipment.
10	The system shall archive all shipment and environmental sensor data for a minimum of 5 years.

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REQ. #	NON-FUNCTIONAL REQUIREMENTS
1	The system shall process and display sensor data within **5 seconds** after it is received.
2	The system shall achieve at least **98% successful data transmission** during transportation.
3	The monitoring platform shall maintain **99% uptime** during active deliveries.
4	All transmitted data shall be encrypted using **TLS encryption**.
5	The system shall restrict access to shipment monitoring dashboards to authenticated and authorized users only.
6	The system shall support monitoring at least **10,000 simultaneous shipments**.
7	The dashboard shall allow users to view shipment status with **no more than 3 clicks**.
8	The system shall automatically reconnect within **30 seconds after network loss.**
9	Sensor data stored in the database shall not be altered without logging the modification history.
10	The system shall allow system updates without interrupting the monitoring service.